Supermarket POS System

1. Course Name

Cap1: System Analysis and Design

2. Team Information

Team Name: Prime Logic Inc.

Name	Group	Student ID	Role
Bobur	I24A	202490377	Leader
Azizjon	I24A	202490017	Project Manager
Ulug'bek	I24A	202490206	Designer
Erkin	I24A	202490286	Software Engineer
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3. Project Title / Description

This project aims to develop a database-driven **Supermarket POS System** using SQL. The system records and manages payment information when a customer purchases a product through a Point of Sale (POS).

It stores key details such as:

- Customer identity
- Purchased products and quantity
- Total amount
- Payment method
- POS location
- Transaction date and time

Each transaction is first recorded with a Pending status and later updated to Confirmed or Failed based on the payment result.

The database uses multiple related tables:

- Customers

- Products
- POS Terminals
- Transactions
- Transaction Items

This system demonstrates how businesses track sales, confirm payments, and maintain reliable transaction records for reporting and analysis.

4. Problem Statement

Businesses need a reliable way to record and verify customer payments made through POS systems. Without a structured system, transaction data may be lost, inconsistent, or difficult to track, causing errors in payment confirmation and financial reporting.

This project solves the problem by creating a structured SQL database that:

- Securely stores transaction details
- Tracks payment status (Pending, Confirmed, Failed)
- Maintains accurate records of who purchased what, when, and where

5. Project Objectives

- Design a structured SQL database for transaction management
- Record customer, product, POS, and time information
- Implement payment status tracking
- Ensure data integrity using relationships and constraints
- Enable easy retrieval of transaction data for monitoring and reporting

6. Scope of the System

The system manages transaction data generated from a POS. It includes:

- Customer information
- Product details
- POS terminal data
- Transaction records with items, amount, payment method, status, and timestamp

Limitations:

- No user interface
- No real payment gateway integration
- No inventory management or authentication
- Designed for educational purposes only

7. Expected Users

- Store Cashiers – record sales transactions
- Store Managers – review sales and transaction history
- Business Owners – analyze revenue and performance

- System Administrators – maintain the database

8. Project Timeline (5 Weeks)

Week	Activities
Week 1 (06 Feb – 13 Feb)	Project planning
Week 2 (13 Feb – 20 Feb)	BPM, Use Case
Week 3 (20 Feb – 27 Feb)	BPM, Use Case
Week 4 (27 Feb – 06 Mar)	ERD Design
Week 5 (06 Mar – 13 Mar)	Report writing & presentation

9. Project Outcomes

- Understand relational database design
- Gain practical SQL experience (CREATE, INSERT, UPDATE, SELECT)
- Learn transaction management and table relationships
- Understand real-world payment and sales processes
- Improve problem-solving and system design skills

10. Conclusion

The project provided a comprehensive introduction to relational database management, beginning with the critical technical task of local SQL environment configuration. By leveraging a collaborative "vibe coding" methodology and integrating AI-driven code generation, the development process was streamlined, allowing for a primary focus on high-level logic and system architecture rather than manual syntax.

Furthermore, proficiency was established in utilizing the **draw.io** (app.diagrams.net) web interface for technical documentation. This involved the formal translation of system requirements into standardized visual models, specifically **Use Case Diagrams**, **Business Process Models (BPM)**, and **Entity-Relationship Diagrams (ERD)**. The synthesis of these tools facilitated a robust understanding of both backend implementation and structured system design.

11. References

Tutorials to use draw.io

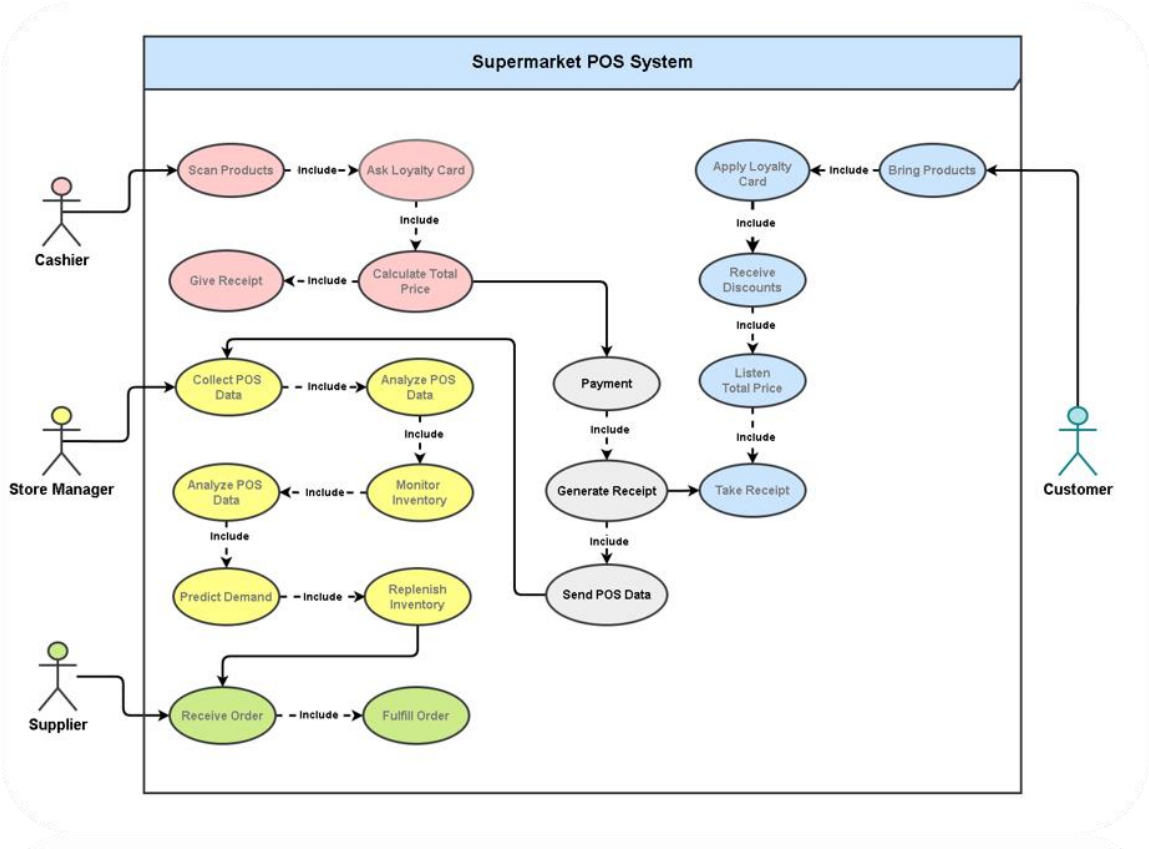
https://youtu.be/1T4igDVMVEg?si=j7No6fdE6kwL_tx4

https://youtu.be/_MzTijsoSe8?si=b-XU5kWIGHU4IU1-

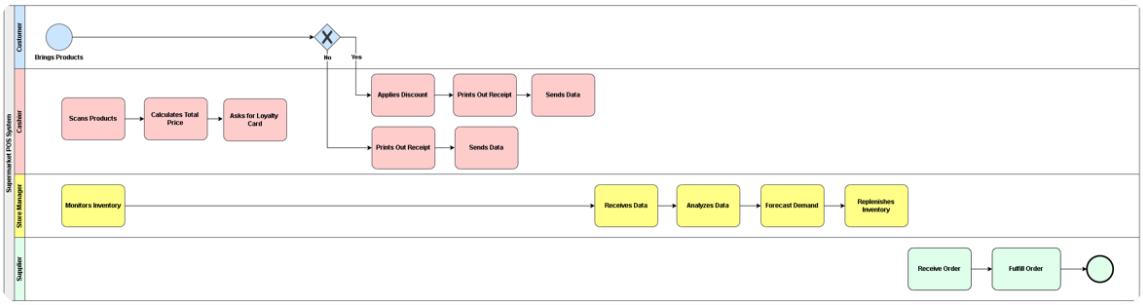
https://youtu.be/JYZPdU5F2iM?si=rdwDp-VCgoB28a1b

Tutorial to install MySQL: https://youtu.be/hiS_mWZmmI0?si=1NjFijY2WsoAqJvd

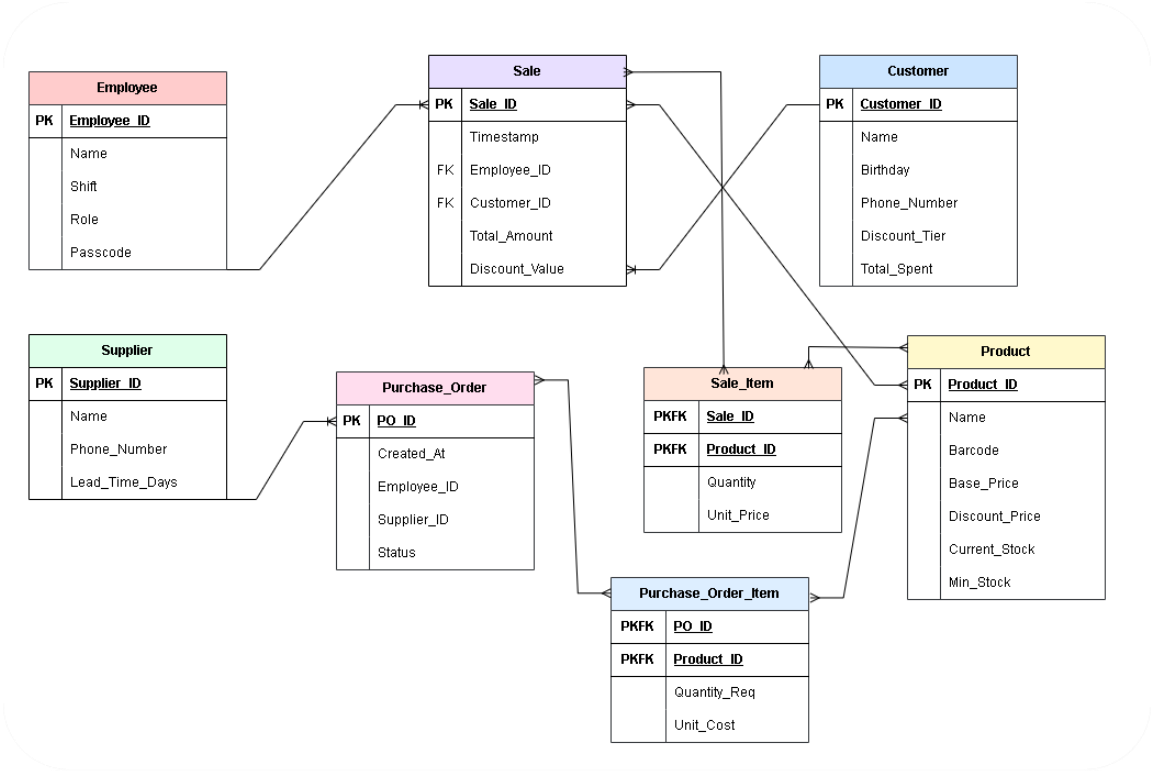
12. Appendix



Use Case



Business Process Model



Entity Relationship Diagram

SQL Code:

```
-- ========================================
-- CREATE DATABASE
-- ========================================

CREATE DATABASE Prime_Logic;

USE Prime_Logic;

-- ========================================
-- 1. INDEPENDENT TABLES
-- ========================================

CREATE TABLE Employee (
    Employee_ID INT AUTO_INCREMENT PRIMARY KEY,
    First_Name VARCHAR(50) NOT NULL,
    Last_Name VARCHAR(50) NOT NULL,
    Role ENUM('Cashier','Manager','Admin') NOT NULL,
    Passcode_Hash VARCHAR(255) NOT NULL
);
```

```
CREATE TABLE Customer (
    Customer_ID INT AUTO_INCREMENT PRIMARY KEY,
    Phone_Number VARCHAR(15) UNIQUE,
    Discount_Tier DECIMAL(3,2) NOT NULL,
    Total_Spent DECIMAL(10,2) DEFAULT 0.00
);

CREATE TABLE Supplier (
    Supplier_ID INT AUTO_INCREMENT PRIMARY KEY,
    Company_Name VARCHAR(100) NOT NULL,
    Contact_Email VARCHAR(100) NOT NULL,
    Lead_Time_Days INT NOT NULL CHECK (Lead_Time_Days >= 0)
);

CREATE TABLE Product (
    Product_ID INT AUTO_INCREMENT PRIMARY KEY,
    Barcode VARCHAR(50) UNIQUE NOT NULL,
    Product_Name VARCHAR(100) NOT NULL,
    Base_Price DECIMAL(10,2) NOT NULL CHECK (Base_Price >= 0),
    Current_Stock INT NOT NULL CHECK (Current_Stock >= 0),
    Min_Threshold INT NOT NULL CHECK (Min_Threshold >= 0)
);

-- ==========================================
-- 2. DEPENDENT TABLES
-- ==========================================

CREATE TABLE Sale (
    Sale_ID INT AUTO_INCREMENT PRIMARY KEY,
    Timestamp DATETIME DEFAULT CURRENT_TIMESTAMP,
    Employee_ID INT NOT NULL,
    Customer_ID INT,
    Total_Amount DECIMAL(10,2) NOT NULL CHECK (Total_Amount >=
0),
    Discount_Value DECIMAL(10,2) DEFAULT 0.00,

    FOREIGN KEY (Employee_ID)
        REFERENCES Employee(Employee_ID)
        ON DELETE RESTRICT
        ON UPDATE CASCADE,

    FOREIGN KEY (Customer_ID)
        REFERENCES Customer(Customer_ID)
        ON DELETE SET NULL
        ON UPDATE CASCADE
```

```
);

CREATE TABLE Purchase_Order (
    PO_ID INT AUTO_INCREMENT PRIMARY KEY,
    Created_At DATETIME DEFAULT CURRENT_TIMESTAMP,
    Employee_ID INT NOT NULL,
    Supplier_ID INT NOT NULL,
    Status ENUM('Pending','Ordered','Received','Cancelled')
NOT NULL,

    FOREIGN KEY (Employee_ID)
        REFERENCES Employee(Employee_ID)
        ON DELETE RESTRICT
        ON UPDATE CASCADE,

    FOREIGN KEY (Supplier_ID)
        REFERENCES Supplier(Supplier_ID)
        ON DELETE RESTRICT
        ON UPDATE CASCADE
);

-- ========================================
-- 3. ASSOCIATIVE TABLES
-- ========================================

CREATE TABLE Sale_Item (
    Sale_ID INT,
    Product_ID INT,
    Quantity INT NOT NULL CHECK (Quantity > 0),
    Unit_Price DECIMAL(10,2) NOT NULL CHECK (Unit_Price >= 0),

    PRIMARY KEY (Sale_ID, Product_ID),

    FOREIGN KEY (Sale_ID)
        REFERENCES Sale(Sale_ID)
        ON DELETE CASCADE
        ON UPDATE CASCADE,

    FOREIGN KEY (Product_ID)
        REFERENCES Product(Product_ID)
        ON DELETE RESTRICT
        ON UPDATE CASCADE
);

CREATE TABLE Purchase_Order_Item (
```

```
    PO_ID INT,
    Product_ID INT,
    Quantity_Req INT NOT NULL CHECK (Quantity_Req > 0),
    Unit_Cost DECIMAL(10,2) NOT NULL CHECK (Unit_Cost >= 0),

    PRIMARY KEY (PO_ID, Product_ID),

    FOREIGN KEY (PO_ID)
        REFERENCES Purchase_Order(PO_ID)
        ON DELETE CASCADE
        ON UPDATE CASCADE,

    FOREIGN KEY (Product_ID)
        REFERENCES Product(Product_ID)
        ON DELETE RESTRICT
        ON UPDATE CASCADE
);
```